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# prolog
# section 005
# Name: Daniel Schlais, drsc232@uky.edu
# Purpose: The program is a dicegame called Twenty-One, 2 players. The players will
take turns rolling die or choosing to pass. The first player to reach or exceed 21
loses.
# Pre-conditions: The users must enter their player names and a response (P or R).
Dice rolls are randomly generated.
# Post-conditions: The program will output each round, what choices were made, the
totals, who lost and who won the game.

# import graphics
from graphics import *
import random
# Checks for point within defined region
def between(p1, p2, p3):
    return (p1.getX() <= p2.getX() <= p3.getX()) and \
           (p1.getY() <= p2.getY() <= p3.getY())
# Sets die image to the anchor point
def d_die(value, win, anchorPoint):
    filename = f"{value}.gif"
    img = Image(anchorPoint, filename)
    img.draw(win)
    return img

# define pass or roll
#purpose: get user choice for turn, pass or roll
#precon: name(parameter, sting) and choice (input)
#postcon: user response (R or P)
def pass_roll(name, win):

    prompt = Text(Point(250, 100), f"{name}: Pass or Roll?")
    prompt.setSize(18)
    prompt.draw(win)

    pass_rec = Rectangle(Point(100, 140), Point(200, 200))
    pass_rec.setFill("lightgray")
    pass_rec.draw(win)
    pass_lab = Text(Point(150, 170), "Pass")
    pass_lab.draw(win)

    roll_rec = Rectangle(Point(300, 140), Point(400, 200))
    roll_rec.setFill("lightgray")
    roll_rec.draw(win)
    roll_lab = Text(Point(350, 170), "Roll")
    roll_lab.draw(win)

    choice = ""
# while input is not "P" and not "R":
    while choice == "":
        click = win.getMouse()
        if between(Point(100, 140), click, Point(200, 200)):
            choice = "P"
        elif between(Point(300, 140), click, Point(400, 200)):
            choice = "R"
#cleanup
    prompt.undraw()
    pass_rec.undraw()
    pass_lab.undraw()

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roll_rec.undraw()
roll_lab.undraw()

return choice

# define play turn
#purpose: to carry out each turn for each player, a player rolls or passes, if
available.
#precon: player name (string), total (int) and passes (int)
#postcon: returns updated totals and passes
def play_turn(name, total, passes, win, die_anchor, pmsg):
    roll_value = 0
    # if passes > 0:
    if passes > 0:
        choice = pass_roll(name, win)
    # if choice == "P"
    if choice == "P":
        passes -= 1
        pmsg.setText(f"{name} passed.")
        die_img = d_die(0, win, die_anchor)
    else:
        roll_value = random.randint(1, 6) # generate random die roll
        total += roll_value # update totals
        pmsg.setText(f"{name} rolled a {roll_value}.")
        die_img = d_die(roll_value, win, die_anchor)
    else:
        roll_value = random.randint(1, 6)
        total += roll_value
        pmsg.setText(f"{name} rolled a {roll_value}.")
        die_img = d_die(roll_value, win, die_anchor)

win.getMouse()

#cleanup
die_img.undraw()
pmsg.setText("")

# return total and passes
return total, passes

# main program
def main():
    # create window
    win = GraphWin("Twenty One", 500, 500)

    head_anchor = Point(250, 50)
    msg_anchor = Point(250, 350)
    p1_anchor = Point(100, 50)
    p2_anchor = Point(400, 50)
    die_anchor = Point(250, 250)

    header = Text(head_anchor, "")
    header.setSize(20)
    header.draw(win)

    p1_t = Text(p1_anchor, "")
    p1_t.setSize(12)
    p1_t.draw(win)
    p2_t = Text(p2_anchor, "")

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p2_t.setSize(12)
p2_t.draw(win)

msg_obj = Text(msg_anchor, "")
msg_obj.setSize(16)
msg_obj.draw(win)

name_l1 = Text(Point(250, 100), "Enter Player 1: ")
name_l1.draw(win)
z1 = Entry(Point(250, 130), 20)
z1.draw(win)

name_l2 = Text(Point(250, 170), "Enter player 2: ")
name_l2.draw(win)
z2 = Entry(Point(250, 200), 20)
z2.draw(win)

msg_obj.setText("Click to continue")
win.getMouse()

player1 = z1.getText()
player2 = z2.getText()

#cleanup
name_l1.undraw()
name_l2.undraw()
z1.undraw()
z2.undraw()
msg_obj.setText("")

# initialize totals
total1 = 0
total2 = 0
passes1 = 3
passes2 = 3
round_n = 1

while total1 < 21 and total2 < 21:

    header.setText(f"Round {round_n}")

    total1, passes1 = play_turn(player1, total1, passes1, win, die_anchor,
msg_obj)

    p1_t.setText(f"{player1}: {total1} (passes {passes1})")
    p2_t.setText(f"{player2}: {total2} (passes {passes2})")

    if total1 < 21:

        total2, passes2 = play_turn(player2, total2, passes2, win, die_anchor,
msg_obj)

        p1_t.setText(f"{player1}: {total1} (passes {passes1})")
        p2_t.setText(f"{player2}: {total2} (passes {passes2})")

        round_n +=1

header.setText("Game Over")

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    if total1 >= 21:
        msg_obj.setText(f"{player2} wins! Click to exit.")
    else:
        msg_obj.setText(f"{player1} wins! Click to exit.")

    win.getMouse()
    win.close()

main() # end main program
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